

1. Hello, this is 6 Minute English from BBC Learning English.

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2. I'm Neil.

我叫尼尔。

3. And I'm Becca.

我叫贝卡。

4. Neil, what do the following animals have in common?

尼尔，以下这些动物有什么共同点呢？

5. Zebras, giant pandas and magpies.

斑马、大熊猫和喜鹊。

6. That's...

那就是.....

7. Pretty easy, Becca.

挺简单的，贝卡。

8. They're all black and white.

它们都是黑白的。

9. Right.

对。

10. But do you know why these animals evolved to be black and white?

但你知道为什么这些动物会进化成黑白相间的外表吗？

11. I'm not so sure.

我不太确定。

12. Probably something about survival.

大概是与生存有关的某种东西吧。

13. That's usually the answer, isn't it?

通常都是这个答案，不是吗？

14. It is.

确实是。

15. The animal world is full of colour, shiny blue peacocks and bright green frogs.

动物世界充满了缤纷的色彩，有闪闪发光的蓝色孔雀，也有鲜绿色的青蛙。

16. But in fact, hundreds of animal species are black and white.

但实际上，有数百种动物是黑白的。

17. In this episode, we'll try to find out why, while also learning some useful new words and phrases.

在这一集中，我们将试图弄清楚原因，同时也会学习一些有用的新单词和短语。

18. And remember, all the vocabulary from this episode is available on our website, bbclearningenglish.com.

请记住，这一集中出现的所有词汇都可以在我们的网站**bbclearningenglish.com**上找到。

19. OK, time for a question for you, Becca.

好的，现在轮到你提问了，贝卡。

20. According to one theory, some animals are black and white to warn potential predators that they're dangerous or they taste bad.

根据一种理论，某些动物之所以呈现黑白相间的颜色，是为了向潜在的捕食者发出警告，表明它们具有危险性或味道不佳。

21. But what is the scientific term for this?

但是，这个现象在科学上叫什么名字呢？

22. Is it a camouflage, b mimicry or c aposomatism?

这是伪装、拟态，还是警戒色作用呢？

23. Well Neil, I'm going to go with c just because I've never heard of that word before.

嗯，尼尔，我选择“c”这个选项，只是因为我之前从未听说过这个词。

24. OK.

OK.

25. I like...

我喜欢.....

26. I like your style.

我喜欢你的风格。

27. Some scientists think zebras' distinctive black and white stripes prevent bites from flies and other insects.

一些科学家认为，斑马身上独特的黑白条纹能够有效防止苍蝇及其他昆虫的叮咬。

28. Zoologist Dr Martin Howe tested this theory by covering horses – a close relative of the zebra – in stripes.

动物学家马丁·豪博士通过给马这种与斑马有密切亲缘关系的动物涂上条纹来验证这一理论。

29. Here Martin discusses his experiment with Caroline Steele from BBC World Service programme CrowdScience.

在这里，马丁与BBC世界服务节目“CrowdScience”的卡罗琳·斯蒂尔一起讨论了他的实验。

30. Do we know what it is about the stripes that's putting them off?

我们知道是条纹的什么特点让它们感到反感吗？

31. Is it somehow like visually confusing?

这会不会在视觉上造成一些混淆呢？

32. We originally set out to test whether maybe there were these optical illusions at play.

我们最初的目的就是试图验证是否确实存在这类视觉错觉现象。

33. So we did a whole bunch of tests where we tried different patterns of rugs but it turned out that practically every high contrast pattern, every black and white pattern we used, almost every pattern was really effective at stopping flies.

因此，我们进行了大量的实验，尝试了各种不同图案的地毯，结果发现，几乎所有高对比度的图案，以及我们使用的所有黑白图案，都非常有效地能阻止苍蝇的靠近。

34. Martin conducted a whole bunch of – meaning a lot of – tests, using different colours and patterns.

马丁进行了大量的测试，使用了不同的颜色和图案。

35. Black and white stripes created an optical illusion – a visual trick that makes your eyes see things differently from how they really are.

黑白条纹会产生一种视觉错觉——这种视觉技巧会让你的眼睛以与事物实际外观不同的方式来看待它们。

36. As a result, the flies were put off.

结果，这些苍蝇都被吓跑了。

37. They lost interest and didn't bite.

他们失去了兴趣，也就不再继续参与了。

38. So zebras' stripes confuse insects, telling them to look elsewhere for food.

因此，斑马的条纹会迷惑昆虫，让它们去其他地方寻找食物。

39. But are there other reasons animals are black and white?

但是，动物之所以呈现黑白相间的颜色，是否还有其他原因呢？

40. Nobody who's seen giant pandas in a zoo would find them hard to spot.

任何在动物园里见过大熊猫的人都会发现，它们非常容易辨认。

41. Against a grassy background they stand out easily.

在绿色的背景映衬下，它们显得格外醒目。

42. But seen from a distance in the snowy mountain forest where they live, black and white makes the perfect disguise.

但从他们居住的那些被雪覆盖的山林中远处看去，黑白相间的色彩反而构成了绝佳的伪装。

43. Here's Professor Tim Carrow, a world authority on animal behaviour, explaining more to Caroline Steele for BBC World Service's CrowdScience.

以下是世界动物行为研究领域的权威专家蒂姆·卡罗教授，他为BBC世界服务台的CrowdScience节目向卡罗琳·斯蒂尔进一步解释相关内容。

44. The giveaway for me was 15 photos of giant pandas taken either fairly close up or at a distance.

对我来说，这份“收获”是15张大熊猫的照片，这些照片或是从很近的距离拍摄的，或是从较远的距离拍摄的。

45. And in two of those photos, I was unable to see anything in the picture other than rocks and snow.

在其中的两张照片里，我只能看到岩石和积雪，其他什么都没有。

46. And then later on, I realized that, in fact, there was a giant panda in the center of that picture some distance away.

后来我才发现，实际上在那张图片的中央，有一只大熊猫，距离我有一段距离。

47. Immediately, that clicked in my mind.

立刻，这个想法在我脑海中浮现了出来。

48. The pandas were so well camouflaged, Tim couldn't spot them in the photos.

那些熊猫伪装得如此巧妙，蒂姆在照片中根本无法发现它们。

49. That's when something clicked in his mind, a phrase used when you suddenly understand something after trying for a long time.

就在那时，他的脑海中突然灵光一现——这就是那种在经过长时间努力后终于恍然大悟时的感觉。

50. Tim understood that pandas use colour to hide.

蒂姆明白，熊猫会利用颜色来进行伪装。

51. He calls this a giveaway, something that makes it easy for people to discover something secret or unknown.

他称这种现象为一种“泄露秘密的行为”，即某种会让人轻易发现某些隐秘或未知事物的行为。

52. So, are animals black and white to repel insects, like zebras, or for camouflage, like pandas?

那么，动物们身上那些黑白相间的图案，是为了像斑马那样驱赶昆虫，还是为了像熊猫那样进行伪装呢？

53. Let's give the final word to Hannah Rowland, a researcher into another black and white animal, penguins.

让我们让另一位研究黑白动物的研究者——汉娜·罗兰德来发表最后意见吧，她研究的对象就是企鹅。

54. Really, science never has a black and white answer.

事实上，科学从来不存在非黑即白的答案。

55. Excuse the pun, with penguins being black and white.

恕我开个双关玩笑——毕竟企鹅可是黑白的啊。

56. It's often a very complex mixture of reasons.

这往往是由多种复杂原因共同作用的结果。

57. The answer to why animals are black and white is not black and white.

对于“为什么动物的毛色是黑色或白色的”这个问题，答案本身并非非黑即白。

58. A situation described as black and white means one where it's easy to understand what's right and wrong.

被描述为非黑即白的情况，意味着人们很容易分辨什么是对什么是错。

59. In fact, there's no one reason why animals are black and white.

事实上，动物之所以呈现黑白相间的颜色，并没有特定的原因。

60. Zebras avoid bites, pandas hide from predators and some animals like skunks warn other animals they're dangerous.

斑马会避开咬伤，熊猫会躲开捕食者，而像臭鼬这样的动物则会向其他生物发出警告，表明自己具有危险性。

61. It all depends on the specific animal.

这完全取决于具体的动物种类。

62. And talking of skunks reminds me of my question, Becca.

说到臭鼬，这就让我想起了我的问题，贝卡。

63. According to one theory, some animals are black and white to warn potential predators they're dangerous or taste bad.

根据一种理论，某些动物之所以呈现黑白相间的颜色，是为了向潜在的捕食者发出警告，表明它们具有危险性或味道不佳。

64. But what is the scientific term?

但是，它的科学术语是什么呢？

65. And I chose C, aposomatism, because I never heard of it before.

而我选择了C选项，即“拟态防御”，因为之前我从未听说过这种机制。

66. Well, that's a great way of choosing and it's done well for you because you're correct.

嗯，这是一种很好的选择方式，而且对你来说效果也不错，因为你的判断是正确的。

67. OK, let's recap the vocabulary we've learned starting with optical illusion – a visual trick where your eyes perceive things differently from how they really are.

好的，让我们回顾一下我们所学过的词汇。首先来看看“视觉错觉”这个词吧——这是一种视觉上的欺骗，会让你的眼睛以与事物实际存在的方式不同的方式来感知它们。

68. A whole bunch of something means a lot of something.

大量的某物意味着有许许多多的某物。

69. If something puts you off, it causes you to dislike it or lose interest.

如果某件事让你感到反感，那么它就会让你对它产生厌恶情绪或失去兴趣。

70. A giveaway is something that makes it easy for people to discover something secret or unknown.

某种线索或提示会帮助人们轻易发现某些隐藏的或未知的事物。

71. The phrase something clicked in my mind is used when you suddenly understand something at last.

“我突然明白了” 这个表达就是用来形容当你终于弄清楚某件事的时候。

72. And finally, if you describe something as black and white, you mean it's easy to understand what is right and wrong.

最后，如果你将某件事描述为非黑即白的，那就意味着人们很容易分辨什么是对什么是错。

73. Once again, our six minutes are up, but head over to our website bbclearningenglish.com for a quiz and worksheet for this episode.

再次提醒，我们的六分钟时间已经结束了。不过，您可以访问我们的网站 bbclearningenglish.com，那里有针对这一集的测验和练习题。

74. See you there soon, but for now it's goodbye.

很快会在那里见到你，但现在先告别了。

75. Goodbye.

再见。

闪电英语